Jason Toma

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EDUCATION

Northeastern University, Boston, MA — GPA: 3.546
Candidate for Bachelor of Science in Mechanical Engineering — May 2027
Relevant Courses: Thermodynamics, Mechanics of Materials, Material Science, Statics, Differential Equations
Activities: Northeastern Electric Racing, Intramural Basketball, Northeastern Spikeball
Awards: Dean's Scholarship

WORK EXPERIENCE

HighRes Biosolutions — Beverly, MA
Mechanical Engineering Co-op — Jan 2025 - June 2025
- Designed and modeled 3D-printed Onyx sensor mount fixtures and adapters for an automated storage system in SolidWorks, incorporating steel drill bushings and brass threaded inserts
- Created dimensioned drawings and bills of material for new parts and released them through engineering change orders in Bluestar PLM, with design review by senior engineers
- Modeled and detailed a machined plate with multiple hole patterns for production and checked fixture fit against full system assemblies

Robert Reiser & Co, Inc. — Canton, MA
Mechanical Engineering Co-op — Jan 2024 - June 2024
- Designed and modeled a variety of product extrusion inserts using SolidWorks that were 3D-printed and tested for customers.
- Created part callouts and dimensional drawings for a variety of parts and assemblies in SolidWorks and organized them in SolidWorks PDM.
- Assembled water wheels and cutoffs for industrial meat processing machines that would be shipped to customers.

ACADEMIC PROJECTS

Capstone Project, Automatic Net Tensioner, Mechanical Engineering — Summer 2026 - Present
- Collaborated with team of 4 mechanical engineering students to design a motorized tensioning system for competition volleyball nets, replacing a manual hand-crank winch
- Measured cable tension with a force gauge and benchmarked commercial winch systems, industry standards, and patents to define load requirements and design constraints
- Developing a motor driven winch with load cell tension feedback in SolidWorks, progressing from 3D-printed prototypes to machined aluminum or steel parts for testing

Carnival Game Project, Cornerstone of Engineering — Spring 2023
- Collaborated with team of 4 interdisciplinary engineering students to create an arduino based game, to display at in-class carnival
- Designed 9 character figures in AutoCAD and used laser cutter to cut from balsa wood to be used as targets for the Carnival Game
- Built wooden stand and backboard, adhering targets, to complete game construction

SKILLS

Computer Applications: SolidWorks (pursuing CSWA), SolidWorks PDM, Bluestar PLM, AutoCAD, Excel, Wrike
Machining: 3D-printing (Grabcad), Laser Cutting

VOLUNTEER EXPERIENCE

Young Men's Service League — San Mateo, CA
Community Service Organization
- Participated in beach clean-ups along Ocean Beach, California
- Tutored 2nd grade student on learning to read for 2 months; tutored 4th grade student in mathematics and geography each week online during quarantine for 4 months

INTERESTS

Interests: Golf, Basketball, Rock Climbing, Spikeball, Adobe Illustrator